

ELECANAL

// --- DATA STRUCTURES ---

// unitDatabase: Hash Map (unordered_map) for O(1) unit lookups

// maintenanceQueue: Queue (queue) for FIFO maintenance handling

// undoStack: Stack (stack) for LIFO undo operations

STRUCT Action:

string type, unitID; float oldValue;

// --- MAIN ENGINE ---

FUNCTION Main():

LoadDataFromCSV() // Initialize Hash Maps

WHILE True:

DisplayMainMenu()

choice = GetInput()

IF choice == "1": AdminPortal()

ELSE IF choice == "2": TenantPortal()

ELSE IF choice == "3": SaveDataToCSV(); BREAK

ELSE: Print "Invalid Selection"

FUNCTION AdminPortal():

IF NOT Authenticate(): RETURN

WHILE AdminActive:

DisplayAdminMenu()

adminChoice = GetInput()

SWITCH adminChoice:

CASE "1": // View Units

PRINT unitDatabase.getAll()

CASE "2": // Process Payment

unitID, amount = GetInput()

undoStack.push(Action("PAYMENT", unitID, unitDatabase[unitID].balance))

unitDatabase[unitID].balance -= amount

CASE "3": // View Maintenance

task = maintenanceQueue.pop() // FIFO

PRINT "Resolving: " + task

CASE "4": // Undo

IF NOT undoStack.isEmpty():

act = undoStack.pop() // LIFO

unitDatabase[act.unitID].balance = act.oldValue

CASE "5": RETURN // Logout

FUNCTION TenantPortal():

IF NOT Authenticate(): RETURN

WHILE TenantActive:

DisplayTenantMenu()

tenantChoice = GetInput()

SWITCH tenantChoice:

CASE "1": PRINT unitDatabase[myID].details

CASE "2": issue = GetInput(); maintenanceQueue.push(issue)

CASE "3": RETURN // Logout

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